Oliver Neal

Technical Director, Shiverbug Studios · Unity (C#)

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PROFILE

Technical Director at Shiverbug Studios, leading the technical side of *Out of Water*, a two-player co-op 3D platformer in Unity 6. A systems programmer at heart: state-machine-driven characters and enemy AI, local co-op and multiplayer plumbing, tools for the team, and the profiling that keeps a scene inside its frame budget.

First Class Honours in BSc (Hons) Games Development from Teesside University and winner of Best Games Programming and Development Artefact at ExpoTees 2026. Now studying an MA in Games Design at Teesside alongside the studio role.

Making games in Unity since 2018, building them consistently since 2023. Nine shipped projects, most playable in the browser, with source on GitHub and technical breakdowns at oliverneal.dev.

EXPERIENCE

Technical Director · Shiverbug Studios — Oct 2026 to present

- Lead the technical side of the studio and sit on its board, alongside the MA. Engineering on *Out of Water*: the character and enemy framework, online co-op alongside splitscreen, level design tools, UI and performance, on a layered codebase with automated tests.
- Own the build (git workflow, pull request review, a tagged playtest build every fortnight), the technical plan for the vertical slice, and the studio's repositories and access.
- Get artists and designers working confidently in Unity and git, so an art-only change never has to wait for a programmer.

Gameplay Programmer (Internship) · Shiverbug Studios — Jul to Sep 2026

- Hired for the character and enemy gameplay code on *Out of Water*, an original-IP two-player co-op 3D platformer collectathon; by the end of the ten weeks I was also the studio's buildmaster.
- Rebuilt both playable characters and the enemies from single-file controllers onto component-driven finite and hierarchical state machines, and built the co-op moveset on top.
- Ran the performance pass: GPU-driven rendering, quality tiers, LOD and culling, measured branch to branch with a profiling tool I built for the team.
- Built the splitscreen plumbing (per-player device routing, cameras, character swapping) and a UI Toolkit front end on a shared token system: HUD, character select, pause, save slots and a controls tutorial.
- Handled presentation and tooling: water, caustics and bubble shaders in Shader Graph and custom URP HLSL, a decoupled FMOD layer the audio team plugs into directly, and the blockout interactables and editor tooling the level designers build puzzles from.
- Designed and shipped the studio's public website, shiverbugstudios.com, alongside gameplay work.

Unity 6 · C# · FSM/HSM · Performance · UI Toolkit · FMOD · Cinemachine · Shader Graph · HLSL

TECHNICAL SKILLS

Languages & engines: C#, Unity 6 (2D/3D), Shader Graph, HLSL, Unreal Engine 5, Python

Gameplay & AI: gameplay programming, systems design, finite & hierarchical state machines, AI navigation (NavMesh), local co-op & splitscreen, Unity Input System

Performance: profiling and frame analysis, time-sliced & multi-rate simulation, GPU-driven rendering, LOD and culling, quality tiers, runtime NavMesh baking

Audio, UI & presentation: FMOD, UI Toolkit, Unity UI, Cinemachine, particle VFX

Multiplayer & services: Netcode for GameObjects, Unity Authentication, Relay, Cloud Save

Tools & workflow: Git and Git LFS, branch workflow on a shared team repo, Visual Studio, JetBrains Rider, Figma, Trello, technical documentation

HIGHLIGHTED SOLO PROJECTS

Cyber Station · Unity 3D · C# · AI navigation · simulation — Apr 2026

- Neon station management game, 86 scripts and ~17k lines solo: hundreds of passengers run off one fixed 10 Hz tick, picking facilities by lowest estimated total delay (queue wait plus real NavMesh walk time, both in seconds), with six live ratings feeding back into the spawn rate.
- Final-year artefact: scored 90/100 and won Best Games Programming and Development Artefact at ExpoTees 2026.

Breakdown: oliverneal.dev/cyber-station.html · Play: olivernealdev.itch.io/cyber-station · Source: github.com/OliverNealDev/CyberStation

Tile Turfer · Unity 2D · C# · procedural generation · enemy AI — Jan 2026

- Turf-war shooter where one territory percentage drives difficulty, fire rate and enemy AI at once, across procedurally generated caves with runtime 2D NavMesh baking.

Breakdown: oliverneal.dev/tile-turfer.html · Play: olivernealdev.itch.io/tile-turfer · Source: github.com/OliverNealDev/TileTurfer

Minimalists · Unity 3D · C# · enemy AI · RTS — Aug 2025

- Node-conquest RTS whose enemy AI gets no cheats and no extra information: it commits through the same percentage interface the player uses, sizing each attack from what the target actually holds, across several distinct AI personalities.

Breakdown: oliverneal.dev/minimalists.html · Play: olivernealdev.itch.io/minimalists · Source: github.com/OliverNealDev/Minimalists

Wanted · Unity 6 · C# · stealth AI · one-week game jam — May 2025

- Top-down stealth survival game built in one week for the Tyne to Game jam: a single detection meter drives three-state officer AI, escalating patrols and helicopter passes on a runtime-baked 2D NavMesh.

Breakdown: oliverneal.dev/wanted.html · Play: olivernealdev.itch.io/wanted · Source: github.com/OliverNealDev/Wanted

Unicellular · Unity · C# · FSM · optimisation — Feb 2025

- Idle simulation stress-tested at 1,000 and 10,000 concurrent entities, kept smooth by a decoupled multi-rate tick and time-sliced proximity AI.

Breakdown: oliverneal.dev/unicellular.html · Play: olivernealdev.itch.io/unicellular

3D Netcode Tank Game · Unity 3D · C# · netcode · multiplayer — Jan 2025

- Server-authoritative multiplayer tank shooter: Relay-hosted sessions, cloud authentication, a decoupled turret aiming rig, and a 64-ray hitscan resolved on the server.

Breakdown: oliverneal.dev/tank-game.html · Source: github.com/OliverNealDev/NetcodeTankGame

Pinnable · Unity 2D · C# · arcade · physics — Mar 2024

- Arcade score-attack catch game with a self-escalating spawn director: hazards unlock on a survival timer, then scale their own spawn weight with score, so difficulty tracks skill rather than a fixed curve.

Breakdown: oliverneal.dev/pinnable.html · Play: olivernealdev.itch.io/pinnable · Source: github.com/OliverNealDev/Pinnable

TEAM PROJECTS

Space Bar Simulator · Unreal Engine 5 · Blueprints · team of 10 — Jan 2026

- Retro-futurist first-person bar simulator for Beta Arcade, Teesside's studio-simulation module: owned the player controller on Enhanced Input, a component-based interaction system behind a Blueprint interface, camera-blended stations and the UI computer terminal.

Breakdown: oliverneal.dev/space-bar.html

Candy Corporate · Unity 2D · small team · pixel art — Apr 2024

- Top-down extraction game inspired by Lethal Company; led level design, tile creation and pixel-art integration.

Play: jurgenwithwings.itch.io/candycorporate

EDUCATION

MA Games Design (with Advanced Practice) · Teesside University · in progress — Sep 2026 to 2028

BSc (Hons) Games Development · Teesside University — Jul 2026

- First Class Honours, 80% overall; final-year project scored 84%, its artefact 90/100.

FdSc Games Technologies · Newcastle College University Centre — Jul 2025

- 2:1 (Year 1 Distinction, Year 2 Merit).

BTEC Level 3 National Foundation Diploma, Creative Digital Media Production · Newcastle College — 2023

- Distinction.